

University of Westminster

School of Computer Science and Engineering

6COSC019C – Cyber Security (2025/2026)

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Assignment Details

Weighting:	50%
Qualifying mark:	40%
Description:	Lab Based Portfolio Answers are based on weekly lab activities.
Learning Outcomes covered in this Assignment:	LO3 Evaluate security architecture and design and provide the means to enhance operation security. LO4 Examine cryptography protocols and vulnerabilities and identify attack vectors to exploit them. LO5 Synthesise emerging trends through engagement and analysis with current research.
Handed Out:	09/02/2026
Due Date:	Tuesday 28/04/2026 at 01:00 PM
Expected deliverables:	Single Report
Method of Submission:	Electronic submission on Turnitin (in PDF format); name your file with your student number and the module code. i.e.: WXXXXXXX_6COSC019W

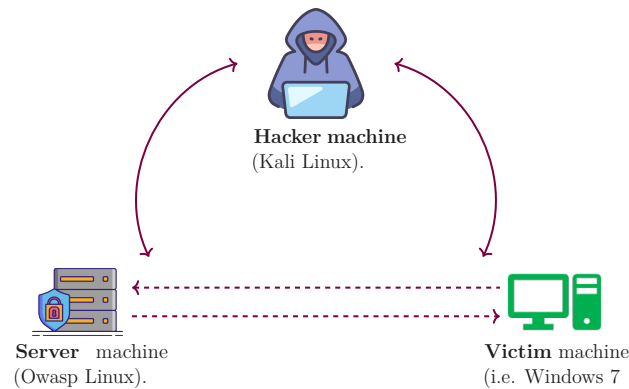
Assessment regulations

See Section 4 of the "Framework for undergraduate course" guide for assessment, penalties, late submission and plagiarism guidance: [Undergraduate Framework](#).

Penalty for Late Submission

Late submission within 24 hours or one working day: **10 marks**, with work in the **40–49%** range capped at **40%**. More than 24 hours or one working day late: **zero mark**, unless a valid Mitigating Circumstances claim is approved.

Coursework description



You must complete all lab activities conducted during the term. Each coursework section is linked to a lab activity.

1. OSINT and Passive reconnaissance

A. During your initial assessment, you conducted an in-depth investigation of the domain name you are testing for penetration on. For this question, you are investigating the Domain name provided for testing for this module cwsenario.uk.

1. Show two DNS enumeration examples you performed. **2 marks** 
2. Explain the potential dangers that the information provided by the examples in the previous question may pose to an organisation. **2 marks** 
two terminal screenshots
3. Suggest one method an organisation can use to minimise the information exposed through DNS analysis activities. **2 marks** 
research (citation)
mitigation - research (citation)

B. One place that penetration testers and hackers alike look for in the early stages of their investigations is the **robots.txt** file for the organisation site.

1. Explain what the robots.txt file is and why it is an important source that hackers investigate. **2 marks** 
2. Show the contents of the **robots.txt** file for the OWASP VM and how you leveraged that information to discover additional findings. **2 marks** 
3. What should an organisation do to mitigate the risk posed by the exposure of information in the robots.txt file? **2 marks** 

2. Active reconnaissance - Scanning and Enumeration

A. During your initial passive reconnaissance, you discovered that the OWASP Vulnerable Machine hosts several different web applications and services.

1. Show the process you followed to discover folders in order to enumerate the running web applications. **1 mark** 
2. Explain why conducting extensive passive reconnaissance beforehand is essential for achieving successful folder enumeration results. **1 mark** 
3. Explain how the folder enumeration tools determine whether a folder exists or not. **1 mark** 

- B. The scanning and enumeration phase of a penetration test usually yields useful information.
1. Show your scanning process to identify hosts while minimising the risk of triggering any firewall or intrusion detection system **2 marks** 
 2. Explain why you chose this specific method to identify hosts. **1 mark** 
 3. Show the results of two different types of TCP scans. For each scan, your results should clearly explain the status of each port, including the reasoning behind its state. **2 marks** 
 4. For each type of the TCP scans you conducted, explain and justify why you chose that specific scan. **2 marks** 
 5. In addition to the open TCP ports you identified, show the open ports that operate using the UDP transport layer protocol. **1 mark** 
- C. While the scanning activity revealed open ports on the OWASP vulnerable machine, it did not provide us with information on the vulnerable machine services nor on what exactly is behind those ports.
1. Show how you enumerated two different services on the OWASP vulnerable network. **2 marks** 
 2. Explain what the Operating system results mean. Compare the results you know with your actual prior knowledge of the operating systems. How accurate the results are. **1 mark** 
 3. Show the enumeration process of the services behind the TCP ports while obfuscating your IP address. **2 marks** 
 4. What measures can the administrators of the OWASP Vulnerable Machine take to reduce the effectiveness of scanning and enumeration, making the results less valuable to a hacker? **1 mark** 
- D. Using basic service enumeration in Nmap can provide useful information about the services running. However, Nmap's scripting engine offers the capability to conduct more in-depth enumeration on specific services.
1. Show how you conducted extensive enumeration on the SSH service using two different scripts. **2 marks** 
 2. Show one IMAP service enumeration activity. **1 mark** 
 3. Which Nmap scripting engine scripts are typically avoided in the early stages of an assessment? **1 mark** 
- E. During the early stages of your scanning, you identified that the Samba service is running on the OWASP Vulnerable Machine.
1. Show how you conducted an extensive enumeration on the Samba service using the enum4linux tool. **2 marks** 
 2. Show two examples of Nmap engine scripts that you could have used to conduct Samba service enumeration instead of using the enum4linux tool. **1 mark** 
 3. Recommend two security measures that could reduce the importance of the Samba service enumeration results. **1 mark** 

3. Exploiting Network Vulnerabilities

- A. Spoofing attacks can be conducted on various layers depending on the access the attacker has prior to the attack.
1. Show how you hijacked the session between the host machine and the OWASP vulnerable machine using the Ettercap tool. **1 mark** 
 2. Show how you conducted a Man-in-the-Middle (MiTM) attack by spoofing the MAC addresses of both the OWASP machine and the host machine using only networking commands. **2 marks** 
 3. Explain what session hijacking is and the threats it poses. **2 marks** 
 4. Explain how an attack can be prevented from conducting a session hijacking attack. **2 marks** 
- B. During the initial scanning and enumeration, you identified several services running on the OWASP vulnerable machine.
1. Show how you enumerated the SSH service using Metasploit. **1 marks** 
 2. Show how you exploited the SSH service using Metasploit to obtain a session shell. **2 mark** 
 3. Explain what privilege escalation is in the context of an SSH session shell. **1 mark** 
 4. Show how you exploited (or attempted to exploit) two other services using Metasploit. **2 marks** 
 5. Explain how you decided which service to exploit and how you selected the appropriate exploitation method. **2 marks** 
- C. Denial of Service (DoS) attacks are one of the most challenging attacks that organisations must deal with continuously.
1. Show two methods you used to carry out a Denial of Service attack on the OWASP vulnerable machine. **2 marks** 
 2. Briefly explain the various methods of conducting Denial of Service attacks and the impact of each. **1 mark** 
 3. Show how DoS attacks can be conducted using the Nmap scripting engine (NSE). **1 mark** 
 4. Briefly explain what organisations can do to mitigate the threats posed by DoS attacks. **1 mark** 

4. Exploiting Web applications Vulnerabilities

- A. During your vulnerability analysis and exploitation phase, you have identified and exploited the SQL injection vulnerability in the DVWA (Damn Vulnerable Web Application).
1. Show how you identified that DVWA is vulnerable to SQL injection and specify the type of database it uses. **2 marks** 

2. Show how you exploited the SQL injection vulnerability by demonstrating the steps of your database enumeration and presenting the most important type of information you were able to obtain. **3 marks** 
 3. Explain what an SQL injection vulnerability is and the threats it poses to an organization. **2 marks** 
 4. Which security principles does SQL injection violate? **1 mark** 
 5. Briefly explain how an organisation can protect its web applications against SQL injection. **1 mark** 
- B. You have identified during a penetration testing activity that the Security Shepherd application is vulnerable to Cross-Site Scripting (XSS).
1. Show how you initially conducted your analysis to identify that the application is vulnerable to XSS. **1 mark** 
 2. During further analysis of the Security Shepherd application, you identified a second XSS vulnerability where input filtering was used. Show how you bypassed the filtering and exploited the vulnerability. **2 marks** 
 3. How dangerous are the threats of XSS attacks to an organisation, and how can a malicious actor leverage the XSS vulnerability to gain confidential information? **2 mark** 
 4. Briefly explain how to protect your web application from Cross-Site Scripting attacks. **2 marks** 
- C. During your vulnerability analysis, you also found and exploited several other vulnerabilities.
1. Show how you conducted Remote File Inclusion (RFI) and website defacement. **2 marks** 
 2. Show how you conducted Cross-Site Request Forgery (CSRF) on the OWASP Virtual machine (DVWA or WebGoat). **1 mark** 
 3. Show how you conducted OS Command Injection on the DVWA web application. **1 mark** 

5. Exploiting Cryptographic weaknesses.

- A. During the exploiting phase you have conducted various cryptanalysis attacks including Brute force and Dictionary attacks.
1. Show how you successfully conducted brute force attack on one service (Web or network service). **2 marks** 
 2. Show how you successfully conducted a dictionary attack on one service (Web or network service) . **2 marks** 
 3. Explain the threats posed by brute force and dictionary attacks, and describe the measures organisations should take to reduce the risk of their environments being compromised by such attacks. **3 marks** 
- B. When exploiting the SQL database of the DVWA application on the vulnerable machine, you managed to acquire the users' passwords in a Hash format.

1. Show how you identified the Hash function algorithm used in the SQL database (you exploited). **2 marks** 
2. Show the hashes for all the SQL users acquired, cracked back to their original plaintext format. **2 marks** 
3. Explain what a **Hash function** is and how it is used in modern web applications? **2 marks** 
4. Suggest one mitigation technique an organisation should implement to prevent hackers from breaking the password hashes of their systems. Justify your answer. **2 mark** 

6. Exploiting Users weaknesses

- A. Many systems are compromised by exploiting user weaknesses, such as luring users to a fake website or service to capture their credentials.
 1. Show how an attacker can lure a normal user of the server to your computer instead of the server machine and show the information obtained. **2 mark** 
 2. Show how you can capture the information obtained from this attack using Wireshark. **1 mark** 
- B. In the previous activity, a social engineering attack was used to capture victim credentials on the OWASP Vulnerable Machine. In many cases, attackers also use social engineering to trick victims into running malicious code to gain further access.
 1. Show the process of crafting the malicious code. **1 mark** 
 2. Show how an attacker would gain initial access once the victim executes the malicious code. **2 marks** 
 3. Identify methods organisations can use to mitigate the threats of social engineering to protect their environments and users. **2 marks** 

Assessment instructions & notes

 Evidence questions (Show): You must complete the lab activity and provide screenshots of outputs, and results. Each screenshot must clearly show your **student ID** in the terminal prompt and include a caption explaining what it shows.

Screenshots must be included within the main body of the report and **not** placed in an appendix.

 Questions with this icon require you to research the relevant weaknesses or exploitation methods and provide a clear, well-structured answer supported by external sources.

 Questions with this icon require you to propose and justify mitigation techniques for the identified threats, supported by external source

Word limit: Maximum 5000 words (excluding references).

Referencing: Harvard referencing required.